

Branching is not a coproduct of branches: unfolding versus descent in the container bifibration

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Abstract

The container bifibration $\mathbf{Cont}(\mathbf{cod}) = \mathbf{Fam}(\mathbf{cod}^{\text{op}})$ is a candidate categorical home for the arboreal / game-comonad analysis of structure and power. In an earlier note we matched, over the base of *paths*, the arboreal relational fibration to a *one-shape* container functor, up to a fibrewise opposite. The obvious way to extend the match from paths to branching trees is to make the branches of a tree into the *shapes* of a container. We show this is wrong, and precisely why: the branch assignment $\Psi(T) = (\text{br}(T), \{\text{chain}_b\})$ computes the colourings of the *branch-unfolding* of the tree — the disjoint union of its root-to-leaf paths — and therefore over-counts the colourings of the tree itself by the exact factor $2^{\sum_v (\lambda(v)-1)}$, where $\lambda(v)$ is the number of branches through the node v . The arboreal fibre reappears inside the container fibre as the sub-poset of *prefix-consistent* colourings: it is the object of descent data for the cover of the tree by its branches, a limit-shaped datum that the shape-coproduct $\mathbf{Fam} = \Sigma$ cannot impose. The faithful encoding uses instead a single shape whose position set is the entire node set; this reproduces the path-level match over the full tree base, the one external input — that the arboreal cartesian lift along a branching subtree embedding is computed by preimage — now discharged from the general cartesian-lift construction of Jakl–Reggio. A separate and higher obstruction — invisibility of k -ary relations for $k \geq 2$ — confines the whole correspondence to the unary fragment. The reusable principle: shared sub-trajectories in a branching process are descent data, not a coproduct over branches.

1 Introduction

The bridge, and the natural guess

Two compositional vocabularies have grown up side by side. On one side, *containers* and *directed containers* — equivalently, small categories [1] — package a system as a set of *shapes* together with, for each shape, a set of *positions*; they organise into the bifibration $\mathbf{Cont}(\mathbf{cod}) = \mathbf{Fam}(\mathbf{cod}^{\text{op}})$ whose fibres are predicates on positions and whose two reindexing legs are direct and inverse image [5]. On the other side, *game comonads* and the *arboreal* axiomatics [3] present model-comparison games (pebble, Ehrenfeucht–Fraïssé, modal) as comonads whose coalgebras are trees of plays, with a forgetful fibration $\mathbb{P}: \mathcal{A} \rightarrow \mathcal{T}$ from coloured trees to trees.

That these two pictures should be *the same picture* is a natural conjecture: both are theories of “a process together with a predicate on its states, pulled back along sub-trajectories”. In an earlier note (Stage 1, [7]) we tested it at the smallest honest scale — a single unary colour, propositionally truncated, over the base $\mathcal{P}$ of *paths* (chains of plays) — and found a clean match, up to a twist. The arboreal path fibration $\mathbb{P}|_{\mathcal{P}}$ is isomorphic to the *fibrewise opposite* of the one-shape container

functor

$$\Phi: \mathcal{P} \rightarrow \mathbf{Cont}(\mathbf{Set}), \quad [n] \mapsto (1, \{[n]\}),$$

with arboreal relation-reflection matching the *opcartesian* (pullback) leg ρ^* of $\mathbf{Cont}(\mathbf{cod})$ — not, as one might first guess, the cartesian leg $(\Sigma_\rho)^{\text{op}}$.

Stage 1 reached only paths, because a one-shape container $(1, \{[n]\})$ is a single chain of positions and a branching tree is not. The registered next step — Stage 2 — conjectured the obvious repair: to reach the full tree base $\mathcal{T}$, make the branches of a tree into the shapes of a container,

$$\Psi(T) = (\text{br}(T), \{\text{chain}_b\}_{b \in \text{br}(T)}),$$

one shape per root-to-leaf path, its positions the nodes along that path. The slogan was “the shape-coproduct $\mathbf{Fam} = \Sigma$ recovers the branch bookkeeping”.

The result

The slogan is backwards, and the way in which it is backwards is the content of this note.

Theorem 1 (Stage 2, the three-way dividing line). *In the unary, propositionally truncated regime, over the base $\mathcal{T}$ of finite rooted trees and rooted-subtree embeddings:*

- (1) (Positive, one shape — proved.) *The arboreal relational fibration extends its Stage 1 match to all of $\mathcal{T}$, branching included, via the one-shape functor $\Psi'(T) = (1, \{\text{nodes}(T)\})$: there is a fibration isomorphism $\mathbb{P} \cong (\Psi'^* \mathbf{Cont}(\mathbf{cod}))^{\text{fib-op}}$ over $\mathcal{T}$. The one external ingredient — that the arboreal cartesian lift along a branching subtree embedding is computed by preimage — is discharged from the general cartesian-lift construction of Jakl–Reggio [4] (Theorem 7.3), which is stated for an arbitrary tree base; see §3.*
- (2) (Negative, branches-as-shapes — proved.) *The conjectured branch functor Ψ does not give the match. Its container fibre is the Boolean lattice of colourings of the branch-unfolding $\widehat{U}(T) = \bigsqcup_b \text{chain}_b$, strictly larger than the arboreal fibre by the factor $2^{\sum_v (\lambda(v)-1)}$ with $\lambda(v)$ the number of branches through v ; the arboreal fibre reappears as the proper sub-poset of prefix-consistent colourings. The excess is nonzero exactly when T branches (first at the three-node V -tree: $8 \subsetneq 16$).*
- (3) (Higher, arity — proved.) *For $k \geq 2$, a k -ary relation is a subset of $\text{nodes}(T)^k$, which is not a subobject in $\mathbf{Set}/\text{nodes}(T)$; the container functor is blind to it and collapses distinct k -ary structures (on two nodes, all 16 binary relations onto the same datum, versus 4 unary). This sits strictly above (1)–(2) and is untouched by the choice of shape functor.*

The three clauses answer three different questions, and conflating them was the error in the original conjecture. Clause (2) is about the *wrong* encoding of branching; clause (1) about the *right* one; clause (3) about a ceiling that no encoding of the unary data can lift.

The mechanism: unfolding versus descent

The heart is clause (2), and it has a one-line diagnosis. The shape-coproduct $\mathbf{Fam} = \Sigma$ is a *coproduct*: applied to the branches of a tree it forms their disjoint union $\widehat{U}(T)$, the universal *unfolding* of the tree by its paths, in which every shared prefix is duplicated once per branch that passes through it. Colouring $\widehat{U}(T)$ freely lets the duplicated copies of a node disagree. The colourings of the *tree* are exactly those that agree across copies — the *prefix-consistent* ones — and reconstructing the

tree from its branch-cover by imposing that agreement is a *gluing*: a descent/equalizer datum, limit-shaped, of the wrong variance for a coproduct. Thus

$\mathbb{P}$ -fibre over T = descent data for the branch-cover $\widehat{U}(T) \rightarrow T$, $\text{Cont}(\text{cod}) \circ \Psi$ -fibre = the raw cover colouring

they coincide iff the cover is trivial, i.e. T is a path — which is exactly why Stage 1 saw no discrepancy. The honest encoding, clause (1), sidesteps the unfolding altogether: keep *one* shape and put the whole node set in the positions, letting the *base* $\mathcal{T}$ carry the tree order through its subtree-embedding morphisms. One copy of each node, no duplication, nothing to glue.

Why it matters

The reusable statement is a modelling warning with teeth. Whenever a branching process — a supply chain forking into parallel routes, an agent-orchestration log splitting into concurrent sub-dialogues, a provenance DAG — is to be encoded in the container bifibration, the temptation is to let each branch be a shape and take their coproduct. Our count shows precisely what that costs: an over-representation by $2^{\sum_v (\lambda(v)-1)}$, one spurious bit of freedom for every extra branch through every shared node. The shared history that distinguishes a tree from its unfolding is *descent data*, and the $\text{Fam} = \Sigma$ coproduct structurally cannot see it. This is the container-side form of our running theme that *composition is a nerve-level-two datum*: the sharing lives one level up from the branches themselves, and is recovered by an explicit gluing step, never read off a product over branches. Clause (3) locates the ceiling of the whole arboreal-container bridge: it is a theorem about the unary fragment, the k -ary relational power of game comonads lying genuinely beyond $\text{Cont}(\text{cod})$.

Outline

Section 2 fixes the two fibrations and recalls the Stage 1 orientation. Section 3 gives the positive one-shape extension (clause 1) and discharges the single external input — the arboreal cartesian lift off paths — from the Jakl–Reggio construction. Section 4 proves clause (2): the exact over-count, the V- and Y-tree witnesses, and the descent diagnosis. Section 5 proves clause (3). Section 6 records scope, open problems, and grant placement.

2 Two fibrations and the Stage 1 match

Throughout we work in the **unary, propositionally truncated** regime: a single relation symbol, interpreted as a *subset* (a subobject, not a proof-relevant family). This is the fragment in which the two theories have a chance of agreeing; §5 shows it is also the ceiling.

2.1 The container bifibration

Recall [1, 2] that a *container* is a pair $(S, \{P_s\}_{s \in S})$ of a set S of shapes and a family of position sets, and that containers organise into $\text{Cont}(\text{cod}) = \text{Fam}(\text{cod}^{\text{op}})$, the bifibration over $\text{Cont}(\text{Set})$ whose fibre over $(S, \{P_s\})$ is $\prod_{s \in S} (\text{Set}/P_s)^{\text{op}}$. We use only its propositional truncation, recorded in [5]:

- **Fibre.** Over $(S, \{P_s\})$ the truncated fibre is $\prod_{s \in S} \text{Sub}(P_s)^{\text{op}} = (2 \sqcup_s P_s, \supseteq)$: predicates on the total position set, ordered by *reverse* inclusion.
- **Cartesian leg.** A container morphism carries a *backward* position map ρ ; its cartesian reindexing is $(\Sigma_\rho)^{\text{op}}$, truncated *direct image* $\rho(-): \text{Sub} \rightarrow \text{Sub}$.

- **Opcartesian (pullback) leg.** The same ρ has an opcartesian reindexing ρ^* , truncated inverse image ρ^{-1} , left adjoint to the cartesian leg.

The two legs are the whole point: “pull a predicate back along a sub-trajectory” is ρ^* , inverse image, *not* the cartesian direction.

2.2 The tree base and the arboreal relational fibration

Definition 2. $\mathcal{T}$ is the category of finite rooted trees of bounded height (a tree is a finite poset with least element root in which every down-set is a chain) and *rooted-subtree embeddings*: a morphism $f: T \rightarrow T'$ is an injection on nodes $j = \text{nodes}(f): \text{nodes}(T) \hookrightarrow \text{nodes}(T')$ whose image is prefix-closed and contains the root. $\mathcal{P} \subset \mathcal{T}$ is the full subcategory of chains (single branches); on $\mathcal{P}$ the only morphisms are initial-segment inclusions — exactly the Stage 1 base.

Definition 3. $\mathcal{A}$ is the category of trees carrying one unary colour $R \subseteq \text{nodes}(T)$ — the unary, truncated relational lift of the arboreal coalgebra category, whose underlying coalgebras are the copresheaves of [6, Lemma 4.1]. The functor $\mathbb{P}: \mathcal{A} \rightarrow \mathcal{T}$ forgets the colour.

Two facts about $\mathbb{P}$ are all we use; both are elementary and recorded in Stage 1.

Lemma 4 (arboreal fibre and reindexing). *1. A vertical map $R \rightarrow R'$ over id_T is a colour-preserving coalgebra homomorphism; it exists iff $R \subseteq R'$. Hence the $\mathbb{P}$ -fibre over T is the full Boolean lattice $(2^{\text{nodes}(T)}, \subseteq)$ — every subset is a legal colouring, branching imposing no constraint.*

*2. Along a subtree embedding $f: T \hookrightarrow T'$ with node injection j , the arboreal reindexing restricts the colour: $f^*R' = R' \cap \text{nodes}(T) = j^{-1}(R')$ (relation-reflection). Inverse image is strictly functorial, so $\mathbb{P}$ is the subobject/predicate fibration $\text{Sub} \rightarrow \mathcal{T}$ with reindexing = pullback.*

The identification in part (1) depends on reading a node of T as the unique play from the root to it: the shared root is a *single* play, so $\mathbb{P}$ colours $\text{nodes}(T)$ with the prefix shared, not the branch-disjoint unfolding. This is the sentence the branch functor Ψ will violate.

2.3 The Stage 1 orientation

Lemma 5 (Stage 1, [7]). *Over the path base $\mathcal{P}$, with $\Phi([n]) = (1, \{[n]\})$: the $\mathbb{P}$ -fibre $(2^{[n]}, \subseteq)$ and the truncated container fibre $\text{Sub}([n])^{\text{op}} = (2^{[n]}, \supseteq)$ are opposite posets; arboreal relation-reflection j^{-1} equals the opcartesian leg ρ^* of $\text{Cont}(\text{cod})$ (and differs from the cartesian $(\Sigma_\rho)^{\text{op}} = \text{direct image}$). Hence $\mathbb{P}|_{\mathcal{P}} \cong (\Phi^* \text{Cont}(\text{cod}))^{\text{fib-op}}$.*

Everything below asks what happens to Lemma 5 off the path base. The fibre-op and the ρ^* identification used only that j is an injection of sets and that inverse image is functorial — never that the positions formed a chain. This is why one of the two candidate extensions survives verbatim, and the other does not.

3 The positive extension: one shape, positions the whole tree

Definition 6. $\Psi': \mathcal{T} \rightarrow \text{Cont}(\text{Set})$ sends T to the *one-shape* container $(1, \{\text{nodes}(T)\})$: a single shape, position set the entire node set as a *flat* set, the tree order discarded by Ψ' and carried by the base $\mathcal{T}$. On a subtree embedding $f: T \hookrightarrow T'$, $\Psi'(f) = (\text{id}_1, \rho_f)$ with $\rho_f: \text{nodes}(T') \twoheadrightarrow \text{nodes}(T)$

the nearest-ancestor retraction (each node of T' to its closest ancestor lying in T ; well-defined as T is prefix-closed and contains the root). On a chain, $\Psi'([n]) = (1, \{[n]\}) = \Phi([n])$, so Ψ' extends Φ ; and Ψ' is functorial, $\rho_{g \circ f} = \rho_f \circ \rho_g$ (nearest ancestor composes through an intermediate subtree).

Proposition 7 (fibre and reindexing match, for every tree). *For every $T \in \mathcal{T}$, branching or not:*

1. *the truncated container fibre over $\Psi'(T) = (1, \{\text{nodes}(T)\})$ is $\text{Sub}(\text{nodes}(T))^{\text{op}} = (2^{\text{nodes}(T)}, \supseteq)$, whose fibrewise opposite is the $\mathbb{P}$ -fibre $(2^{\text{nodes}(T)}, \subseteq)$ of Lemma 4(1), on the nose (one shape, one copy of each node: no over-count);*
2. *relation-reflection j^{-1} along a subtree embedding is the opcartesian leg ρ^* of $\text{Cont}(\text{cod})$ for the arrow carrying j backward, exactly as in Lemma 5, the identification being insensitive to any order on the positions.*

Proof. (1) is immediate: $\bigsqcup_s P_s = \text{nodes}(T)$ since there is one shape, and the Stage 1 fibre-op argument (Lemma 5) referenced only the position *set*. (2) is the Stage 1 argument of Lemma 5, clause by clause, with $[n]$ replaced by the flat set $\text{nodes}(T)$: j^{-1} is functorial inverse image, which is the opcartesian leg of $\text{Cont}(\text{cod})$ along the backward map j . No step used the chain order. $\square$

Proposition 7 matches the two fibrations *fibrewise and on reindexing*. To promote it to an isomorphism of fibrations one needs that the arboreal *cartesian lifts* along subtree embeddings are computed by this same reindexing. This is the single remaining ingredient, and we now establish it.

Lemma 8 (the branching cartesian lift is preimage). *In the unary truncated arboreal fibration $\mathbb{P}$, the cartesian lift of a subtree embedding $f: T \hookrightarrow T'$ (node-injection j) at a colouring R' is $(T, j^{-1}R')$ — preimage — for every T , branching included.*

Proof. This is the specialisation to subtree embeddings of the general cartesian-lift construction of [4], whose Theorem 7.3 is proved for an *arbitrary* tree base, not only for paths, and establishes two facts we use. First, it characterises the cartesian morphisms as the *pathwise embeddings*: f is cartesian iff $f \circ m$ is an embedding for every path embedding $m: P \rightarrow X$. The adjective “pathwise” governs how the test is applied — one path at a time — and places no linearity hypothesis on the domain; a rooted-subtree embedding, being a chain embedding along each branch, is therefore pathwise, hence cartesian. Second, it shows $\mathbb{P}$ is a Street fibration, and its proof builds the lift of a forest morphism $h: T \rightarrow \mathbb{P}Y$ explicitly as the colimit $X = \text{colim}_{i \in T} P_i$ of the representing path embeddings, glued along the order of T . Specialise to $Y = (T', R')$ and $h = j$: each $i \in T$ is represented by the path $P_i = \{\text{root}, \dots, j(i)\}$ carrying the colour restricted from R' (the arboreal substructure embeddings are the regular monos, so a path embedding carries the *induced* colour), and node i is coloured in P_i iff $j(i) \in R'$. Because R' is one fixed subset of T' and $j(i)$ one fixed node, *every* occurrence of i across the diagram carries the identical truth value $[j(i) \in R']$; the colimit therefore colours i by exactly this value, i.e. $\text{colour}(X) = \{i : j(i) \in R'\} = j^{-1}R'$, with underlying tree T (a tree is the colimit of its own paths). Branching is harmless precisely here: two branches sharing a prefix must agree on the colour of a shared node, and they do automatically, since that colour is the single global truth value $[j(i) \in R']$, read off one fixed R' regardless of the branch approached along. (This is exactly the consistency that clause (2)’s Ψ destroys by splitting the branches into separate shapes; §4.) $\square$

Proposition 9 (one-shape extension). *In the unary truncated regime there is an isomorphism of fibrations over $\mathcal{T}$,*

$$\mathbb{P} \cong (\Psi'^* \text{Cont}(\text{cod}))^{\text{fib-op}},$$

so the branching of T is encoded entirely in the base and its subtree-embedding morphisms, while the container functor keeps one flat position set and the opcartesian leg performs the reflection.

Proof. Proposition 7(1) gives the fibrewise anti-isomorphism and (2) the agreement of reindexing; Lemma 8 gives the agreement of cartesian lifts — the arboreal cartesian lift $R' \mapsto j^{-1}R'$ is the preimage that, under the fibrewise opposite, is the opcartesian leg ρ^* of $\mathbf{Cont}(\mathrm{cod})$. Composing the three, $\mathbb{P} \cong (\Psi'^* \mathbf{Cont}(\mathrm{cod}))^{\mathrm{fib-op}}$. $\square$

Remark 10 (what was formerly conditional). An earlier version of this note left Proposition 9 conditional on the property $(*)$ that the $\mathbb{P}$ -cartesian morphisms are the pathwise embeddings at subtree as well as at path embeddings, flagging the non-path case as an input taken from the arboreal literature rather than verified. Lemma 8 discharges it: the non-path case is not a separate hypothesis but the general content of [4], Theorem 7.3, whose “pathwise embedding” tests the embedding condition path-by-path while ranging over all objects, branching trees included. The discharge is confirmed computationally on the branching V- and Y-trees: over every subtree embedding and every colouring R' , the pathwise-embedding lift and the Street-fibration colimit colour both equal $j^{-1}R'$ node by node, consistently on shared prefixes.

4 Branches as shapes unfolds the tree

This section proves clause (2) of Theorem 1: the conjectured branch functor computes the colourings of the wrong object. Nothing here is conditional.

Definition 11. $\Psi: \mathcal{T} \rightarrow \mathbf{Cont}(\mathbf{Set})$ sends T to $(\mathrm{br}(T), \{\mathrm{chain}_b\}_b)$: one shape per root-to-leaf branch b , with position set over b the chain of nodes along b . On a chain there is one branch and $\Psi = \Phi = \Psi'$, so Stage 1 holds; off $\mathcal{P}$ they diverge.

Definition 12. The *branch-unfolding* of T is $\widehat{U}(T) = \bigsqcup_{b \in \mathrm{br}(T)} \mathrm{chain}_b$, the disjoint union of the branch-chains, with *gluing map* $\pi: \widehat{U}(T) \twoheadrightarrow \mathrm{nodes}(T)$, $(b, v) \mapsto v$. For a node v , write $\lambda(v)$ for the number of branches through v (equivalently, the number of leaves above v), so $\pi^{-1}(v)$ has size $\lambda(v)$.

Lemma 13 (the exact over-count). *The truncated container fibre over $\Psi(T)$ is $\prod_b \mathrm{Sub}(\mathrm{chain}_b)^{\mathrm{op}} = (2^{\widehat{U}(T)}, \supseteq)$. Its cardinality is $2^{|\widehat{U}(T)|}$ with*

$$|\widehat{U}(T)| = \sum_b |\mathrm{chain}_b| = \sum_{v \in \mathrm{nodes}(T)} \lambda(v), \quad \text{so} \quad |\widehat{U}(T)| - |\mathrm{nodes}(T)| = \sum_v (\lambda(v) - 1).$$

The pullback $\pi^: 2^{\mathrm{nodes}(T)} \hookrightarrow 2^{\widehat{U}(T)}$ is injective with image the prefix-consistent colourings — those constant on each fibre $\pi^{-1}(v)$. Hence the $\mathbb{P}$ -fibre $(2^{\mathrm{nodes}(T)}, \subseteq)$ embeds as the proper sub-poset of prefix-consistent colourings inside the $\mathbf{Cont}(\mathrm{cod}) \circ \Psi$ -fibre $(2^{\widehat{U}(T)}, \supseteq)$, the excess being the factor $2^{\sum_v (\lambda(v) - 1)}$. The inclusion is proper exactly when some node has $\lambda(v) \geq 2$, i.e. exactly when T branches.*

Proof. The fibre formula is the container fibre of §2 with $\bigsqcup_s P_s = \bigsqcup_b \mathrm{chain}_b = \widehat{U}(T)$. Counting positions by the node they glue to gives $|\widehat{U}(T)| = \sum_v \lambda(v)$. A subset $A \subseteq \widehat{U}(T)$ equals $\pi^*(B)$ for some $B \subseteq \mathrm{nodes}(T)$ iff A is a union of fibres of π , i.e. constant on each $\pi^{-1}(v)$; such B is then unique ($B = \{v : \pi^{-1}(v) \subseteq A\}$), so π^* is injective with the stated image. The complementary degrees of freedom number $\prod_v 2^{\lambda(v) - 1} = 2^{\sum_v (\lambda(v) - 1)}$ (for each v , a consistent colouring fixes all $\lambda(v)$ copies together; a free one chooses them independently). This exponent is positive iff some $\lambda(v) \geq 2$. $\square$

In particular the two fibres have different cardinalities as soon as T branches, so there is *no* fibrewise equivalence $\mathbb{P} \cong (\Psi^* \text{Cont}(\text{cod}))^{\text{op}}$ — not even onto the image of Ψ .

Example 14 (the V-tree). Let V have root r and two leaves a, b . Then $|\text{nodes}| = 3$, the two branches are ra and rb , $|\widehat{U}| = 4$, and $\lambda(r) = 2$, $\lambda(a) = \lambda(b) = 1$, so $\sum_v (\lambda(v) - 1) = 1$. The arboreal fibre has $2^3 = 8$ colourings; the $\text{Cont}(\text{cod}) \circ \Psi$ -fibre has $2^4 = 16$. The 8 excess colourings are precisely the *inconsistent* ones — those colouring the root in branch ra but not in branch rb , or vice versa. The map π identifies the two copies (ra, r) and (rb, r) of the root; consistency is agreement on that single overlap. (Verified by enumeration; `scratch/crown_stage2_witness.py`.)

Example 15 (the Y-tree: deeper sharing). Let Y have root r , a single child c , and c in turn with two children x, y . Then $|\text{nodes}| = 4$, the branches are rcx and rcy , $|\widehat{U}| = 6$, and $\lambda(r) = \lambda(c) = 2$, $\lambda(x) = \lambda(y) = 1$, so $\sum_v (\lambda(v) - 1) = 2$: the whole shared spine $\{r, c\}$ is double-counted. The arboreal fibre has $2^4 = 16$ colourings, the $\text{Cont}(\text{cod}) \circ \Psi$ -fibre $2^6 = 64$, ratio $2^2 = 4$. (Verified; `scratch/crown_stage2_more.py`.)

4.1 The diagnosis: $\text{Fam} = \Sigma$ is unfolding, gluing is descent

Lemma 13 is a count; here is what it means. The branch family $\{\text{chain}_b \hookrightarrow T\}_b$ is a *cover* of T by its paths. Its nerve is governed by the overlaps $\text{chain}_b \times_T \text{chain}_{b'} = \text{chain}_{b \wedge b'}$, the shared prefix down to the meet of b and b' . A colouring of the tree is the same as a family of colourings of the branches that *agree on overlaps*: a *descent datum* for the cover.

Proposition 16 (descent description of the arboreal fibre). *For every T , the image of π^* — the prefix-consistent colourings — equals the set of colourings of $\widehat{U}(T)$ that agree on all pairwise overlaps of the branch cover; and this equals $2^{\text{nodes}(T)}$, the full arboreal fibre. Thus*

$\mathbb{P}$ -fibre over T = descent data (equalizer) for $\widehat{U}(T) \rightarrow T$, $\text{Cont}(\text{cod}) \circ \Psi$ -fibre = the raw, pre-descent cover coinciding iff the cover is trivial, i.e. T a path.

Proof. Two copies (b, v) and (b', v) of the same node v lie in the overlap $\text{chain}_{b \wedge b'}$ precisely when $v \leq b \wedge b'$, i.e. when both b, b' pass through v ; as b, b' range over branches through v these copies exhaust $\pi^{-1}(v)$. Agreement on all overlaps therefore says exactly that a colouring is constant on each $\pi^{-1}(v)$ — prefix-consistency. By Lemma 13 these are the image of π^* , in bijection with $2^{\text{nodes}(T)}$. (The gluing is exact: the tree carries its Alexandrov topology, the branch cover is a cover for it, and a sheaf of colourings glues uniquely — here at the elementary level of a single equalizer, verified in `scratch/crown_stage2_witness.py`: `consistent = agree-on-overlap = 2^{\text{nodes}}`.) $\square$

The conceptual point is a mismatch of variance. $\text{Fam} = \Sigma$ is a coproduct over shapes; fed the branches it yields the disjoint union $\widehat{U}(T)$, the étale unfolding of the tree. Reconstructing the tree is the *opposite* operation — a gluing that *identifies* shared prefixes, a coequalizer/descent equalizer, a limit-flavoured datum. A coproduct cannot impose it. So Ψ realises $\mathbb{P}$ only as the *descent completion* of $\text{Cont}(\text{cod}) \circ \Psi$: an equivalence onto the descent sub-fibration of prefix-consistent colourings, never onto the whole thing.

Remark 17 (reconciliation with clause (1)). Ψ and Ψ' differ exactly by unfolding versus gluing: $\Psi(T)$ is the branch cover, $\Psi'(T)$ is the tree. $\text{Cont}(\text{cod})$ colours whatever flat position set it is handed. Hand it the unfolded cover (Ψ) and it colours the cover, which descend to the tree only after an equalizer Fam cannot supply; hand it the node set (Ψ') and it colours the tree directly. **Branching is not a coproduct of branches:** the faithful multi-shape encoding is a single shape whose position set is the branching node set, with the base $\mathcal{T}$ carrying the order.

5 The arity obstruction is independent and higher

Everything above is $k = 1$: one unary colour. Clause (3) concerns $k \geq 2$ and is a different, strictly higher obstruction — it survives *either* choice of shape functor.

Proposition 18 (non-fullness for $k \geq 2$). *A k -ary relation on T is a subset of $\text{nodes}(T)^k$. For $k \geq 2$ this is not a subobject in $\mathbf{Set}/\text{nodes}(T)$, and $\mathbf{Cont}(\text{cod})$ over $\Psi'(T) = (1, \{\text{nodes}(T)\})$ has fibre $\text{Sub}(\text{nodes}(T))^{\text{op}}$ — unary predicates only. Hence the container fibre (2^n on n nodes) cannot surject onto the arboreal k -ary fibre (2^{n^k} structures): the functor factors through the unary reduct and collapses distinct k -ary structures.*

Proof. $\mathbf{Set}/\text{nodes}(T)$ has as objects maps into $\text{nodes}(T)$; its subobjects are subsets of $\text{nodes}(T)$, not of $\text{nodes}(T)^k$. A k -ary relation determines at most its unary projections, so any two k -ary structures with equal unary reducts — e.g. on two nodes $\{0, 1\}$ the relations $\emptyset$ and $\{(0, 1)\}$, among all $2^4 = 16$ binary relations — map to the same container datum, of which there are only $2^2 = 4$. (Verified; `scratch/crown_stage2_positive.py`) $\square$

This is the same “the forgetful functor is blind to the higher datum” mechanism we have met as out-degree invisibility, as the nerve-level gap $N_1 \subsetneq N_2$, and as composition-being-a-degree-two datum. It sits strictly above the $k = 1$ unfolding/descent discussion of §4 and is untouched by replacing Ψ with Ψ' . The arboreal-container bridge, honestly, is a theorem about the *unary* fragment.

6 Scope, open problems, and placement

What is proved. All three clauses of Theorem 1 are *proved*. Clauses (2) and (3): the exact over-count $2^{\hat{U}} \supsetneq 2^{\text{nodes}}$ with factor $2^{\sum_v (\lambda(v)-1)}$ and its descent reading (Lemmas 13, Prop. 16, Examples 14, 15), and the arity collapse $2^{n^k} \supsetneq 2^n$ (Prop. 18). Clause (1) — the positive one-shape extension over $\mathcal{T}$ — is now proved outright (Prop. 9): the one ingredient that an earlier version left conditional, that the arboreal cartesian lift along a branching subtree embedding is preimage, is discharged in Lemma 8 as the specialisation to subtree embeddings of the general construction of [4], whose Theorem 7.3 holds over an arbitrary tree base.

Open problems.

1. **Proof-relevance.** The match is propositional. The full container fibre $(\mathbf{Set}/\text{nodes})^{\text{op}}$ is strictly richer than $\text{Sub}(\text{nodes})^{\text{op}}$; a proof-relevant comparison would need an arboreal theory with witness-set colours.
2. **The descent completion as a fibration.** Proposition 16 is elementary (a single equalizer per tree). A clean statement “ $\mathbb{P}$ is the descent completion of $\mathbf{Cont}(\text{cod}) \circ \Psi$ as fibrations over $\mathcal{T}$ ” is sketched, not proved here.
3. **Beyond embeddings.** $\mathcal{T}$ here has subtree embeddings only; open maps and non-monic tree morphisms are untreated, inherited from the Stage 1 conventions.

Placement. The net correction to the registered Stage 2 conjecture is clean: it is *not* “ Ψ positive on the unary fragment, failing at $k \geq 2$ ”. It is that the branch-multishape Ψ fails already at $k = 1$ *branching* (unfolding versus descent); the positive Stage 2 holds over the full tree base via the one-shape $\Psi'(T) = (1, \{\text{nodes}(T)\})$; and $k \geq 2$ is a separate, higher obstruction. The reusable principle, and the one worth carrying into the applied side of the programme: *when encoding a branching process in the container bifibration, do not make the branches into shapes*. That unfolds the process and over-counts shared history by one bit per extra branch through each shared node; keep one shape and put the whole state set in the positions, because the sharing — shared provenance, shared sub-trajectory, composition itself — is a descent datum that the $\text{Fam} = \Sigma$ coproduct structurally cannot see. This is the container-side statement of “composition is a degree-two datum”, and it applies verbatim to supply-chain provenance models and to agent-orchestration logs: shared sub-trajectories must be glued by an explicit descent step, not recovered from a product over branches.

Formalisation note. The counts of §4–5 are small finite enumerations (`scratch/crown_stage2_{witness,more}`), and are natural candidates for a Lean oracle in the style of the nerve-level and gerbe engines; the registry node `crown-stage2-multishape-branch-extension` should be created and graded *computed* to match this note. These are lean/registry tasks, flagged here, not done here.

References

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- [7] MacBeth, *Refined Bridge-3 crown, Stage 1: the orientation gap — relation-reflection is ρ^* , not $(\Sigma\rho)^{\text{op}}$* , Kodamai internal proof note, 2 Oct 2026.